# WEEKLY INTERNSHIP REPORT

*Summer Research Programme 2025*

| | |
|---|---|
| **Week Number**<br>Week 4 of 8 | **Report Period**<br>21 / 06 / 2026 → 27 / 06 / 2025 |
| **Group & Project**<br>Group 4 \| Project Title: Event-Driven 6G Scheduling using SimPy + Sionna | **Mentor Name**<br>Dr. Ch Anil Carie |
| **Intern Name(s)**<br>1. Snehitha (AP24110011008)<br>2. Shahistha Anjum (AP24110010993)<br>3. Shanmukh (AP24110011010)<br>4. Sandeep (AP24110010992) | **This week's title**<br>Implementation and Performance Evaluation of Game-Theoretic Defender Models |

**HOW THIS REPORT BUILDS YOUR CONFERENCE PAPER**

Section 2 → Related Work (papers you read + Coursera/YouTube concepts learned)

Section 4b → Results section data (every number here becomes a table row in the paper)

Section 5 → Draft sentences copied directly into the paper manuscript

Section 6 → Figures described here become the paper's figures with captions

Section 8 → Limitations section and Future Work paragraph

Fill every section honestly. A blank section = missing paper content.

## 1 WEEK GOAL

Implementation and Performance Evaluation of Game-Theoretic Defender Models

**This week's goal:** The goal of this week was to continue implementing the game-theoretic defender models introduced in Week 3. The focus was on improving the Stackelberg and Nash Defender modules, testing them using small-scale simulations, and evaluating their performance using metrics such as detection rate and compromised ratio. Additional attention was given to understanding how different payoff values influence defender strategies.

**Key deliverable (the one output that proves the goal was met):**
Completed the basic implementation of Stackelberg and Nash Defender models, performed preliminary performance evaluation, and generated initial comparison results.

## 2 LEARNING — COURSERA + YOUTUBE

### 2A. Coursera Progress

| Course Name | Modules Completed This Week | Score / Status |
|---|---|---|
| Game Theory Applications | Cybersecurity Decision Models | Completed |
| Cybersecurity Decision Models | Defender Resource Allocation, Strategy Comparison | Completed |
| Performance Evaluation | Detection Rate, Compromised Ratio, Risk Score Analysis | Completed |
| Python Data Visualization | Performance Comparison Graphs | In Progress |

### 2B. YouTube Watched This Week

***List every video watched. In the 'Key Takeaway' column write what you actually learned — not just the topic.***

| Video Title | Channel | Duration | Key Takeaway (1 sentence) |
|---|---|---|---|
| Stackelberg Game Applications | Neso Academy | 28 min | Learned how leader-follower strategies improve decision making. |
| Nash Equilibrium Examples | Khan Academy | 18 min | Understood practical examples of equilibrium-based strategies. |
| Python Data Visualization | Corey Schafer | 30 min | Learned to visualize performance metrics using Python. |

### 2C. New Concept Learned (explain in your own words)

This week I learned how performance metrics help evaluate the effectiveness of different cybersecurity strategies. Detection rate and compromised ratio provide a simple way to compare defender models under different attack scenarios. I also understood that the effectiveness of a strategy depends on both the payoff values and the attacker's behavior. These concepts helped me understand how game theory can improve network security decisions.

## 3 DAILY TASK LOG

*Fill this DAILY — not on Friday from memory. One row per day.*

| Day | Task Completed | File / Output Committed to GitHub | Blocker (if any) |
|---|---|---|---|
| Mon | Reviewed Stackelberg and Nash Game concepts. | game_theory_review.md | None |
| Tue | Improved Stackelberg Defender implementation. | stackelberg_defender.py | Required additional testing. |
| Wed | Improved Nash Defender implementation. | nash_defender.py | Strategy comparison required more validation. |
| Thu | Performed small-scale simulations and calculated performance metrics. | simulation_results.csv | Large-scale simulations not completed. |
| Fri | Generated comparison results and documented observations. | performance_report.md | Visualization work partially completed. |

### Weekly Commit Summary

| GitHub Repo URL | Total Commits This Week | Most Important Commit (message) |
|---|---|---|
| https://github.com/Snehithaa14/RESEARCH-INTERN | 6 | Completed basic Stackelberg and Nash Defender implementation with preliminary performance evaluation. |

## 4 TECHNICAL WORK THIS WEEK

### 4A. What I Built / Implemented

This week I improved the preliminary Stackelberg and Nash Defender models developed during the previous week. Small-scale simulations were performed to evaluate both strategies using detection rate and compromised ratio. Basic performance comparison was carried out to understand how different payoff values affect defender decisions. Although the implementation is still under development, the results obtained provide useful insights for future large-scale experiments.

## 4B. Results & Numbers ★ MOST IMPORTANT ★

**WHY THIS SECTION MATTERS**

Every number you record here is a candidate row in your paper's results table.

RULE: If you ran an experiment this week, it MUST appear here as a number. No number = experiment not done.

These rows will be copied directly into the paper in Weeks 7–8.

| Method / Model / Config | Metric 1 (name) | Metric 2 (name) | Metric 3 (name) | Notes |
|---|---|---|---|---|
| Stackelberg Defender | Detection Rate = 85% | Compromised Ratio = 0.17 | Average Risk Score = 0.34 | Improved after strategy refinement. |
| Nash Defender | Detection Rate = 79% | Compromised Ratio = 0.23 | Average Risk Score = 0.41 | Stable performance. |
| Performance Comparison | Models Compared = 2 | Metrics Evaluated = 3 | Simulation Runs = 20 | Initial evaluation completed. |

**Best result this week:** The Stackelberg Defender achieved the highest detection rate while maintaining the lowest compromised ratio among the tested strategies.

**Compared to baseline:** Compared with the preliminary implementation from Week 3, the refined defender models produced slightly better detection performance and lower network risk.

## 4C. Key Figure / Plot This Week

file path: results/strategy_comparison.png

**Caption draft:** Figure 1 compares the performance of the Stackelberg Defender and Nash Defender using detection rate and compromised ratio. The preliminary results indicate that the Stackelberg strategy provides slightly better defensive performance during small-scale simulations.

## 5 PAPER CONTRIBUTION LOG

**THIS SECTION IS THE HEART OF THE REPORT**

Write these sentences THE SAME DAY as the experiment — not on Friday. Memory fades.

These sentences are copied into the paper in Weeks 7–8. The better you write here, the easier the paper sprint.

Every week must have at least 3 candidate sentences. Aim for 5.

## 5A. Candidate Sentences for the Paper

*Format: [Section it belongs in] → [Draft sentence with numbers if possible]*

| Paper Section | Draft Sentence (write as it would appear in the paper) |
|---|---|
| Introduction | Game-theoretic defender strategies provide an effective approach for improving cybersecurity decision making in IoT networks. |
| Related Work | Stackelberg and Nash Game models have been widely adopted for strategic resource allocation and attack mitigation. |
| Methodology | The proposed framework evaluates Stackelberg and Nash Defender strategies using payoff-based decision making. |
| Methodology | Preliminary experiments showed that the Stackelberg Defender achieved a higher detection rate than the Nash Defender. |
| Results | Performance metrics including detection rate and compromised ratio were used to compare the effectiveness of both strategies. |

## 5B. Gap / Limitation Found This Week

The current implementation was evaluated only through small-scale simulations. Larger network topologies and more attack scenarios are required to validate the effectiveness of the proposed strategies. Additional visualization and detailed performance analysis will be carried out in future work.

## 5C. Reference Found This Week

| # | IEEE Citation (Author, Title, Journal/Conf, Year) | Relevance to Our Work |
|---|---|---|
| 1 | M. J. Osborne, *An Introduction to Game Theory*, Oxford University Press, 2004. | Understanding Nash Equilibrium and strategic decision making. |
| 2 | T. Başar and G. J. Olsder, Dynamic Noncooperative Game Theory, SIAM, 1999. | Foundation for Stackelberg Game implementation. |
| 3 | K. Hausken and J. M. Zhuang, Game Theory and Cyber Security, Springer, 2011. | Application of game theory in cybersecurity and attacker-defender modeling. |

## 6 NEXT WEEK PLAN

*Plan this before you leave on Friday. Specific tasks only — not 'continue work on the model'.*

| Day | Planned Task (specific) | Expected Output / Commit |
|---|---|---|
| Mon | Complete performance visualization and generate comparison graphs. | strategy_comparison.png |
| Tue | Perform larger-scale simulations using different attack scenarios. | simulation_results.csv |

| Day | Planned Task (specific) | Expected Output / Commit |
|---|---|---|
| Wed | Analyze performance metrics and compare defender strategies. | performance_analysis.md |
| Thu | Improve payoff calculations and optimize strategy selection. | updated_payoff_engine.py |
| Fri (demo) | Present the overall performance comparison and summarize key observations. | final_performance_report.pdf |

## Coursera + YouTube Plan for Next Week

| | | |
|---|---|---|
| YouTube | Advanced Python Data Visualization | Monday |
| YouTube | Game Theory in Network Security | Wednesday |
| YouTube | Cybersecurity Performance Evaluation | Thursday |

## 7 SELF-ASSESSMENT

**WHY THIS SECTION EXISTS**

These two questions build the skills that separate researchers from coders.

Q1 answers become the 'Surprising Finding' paragraph in the paper's Discussion section.

Q2 answers become better experimental design next week — and better research judgment overall.

Answer honestly. 3–5 sentences each. This is for your growth, not for marking.

### Q1: What surprised you most technically this week?

I was surprised that even small modifications in payoff values noticeably changed the defender's decisions. This demonstrated that strategy optimization depends heavily on the design of the payoff function. The comparison also showed that the Stackelberg Defender consistently performed slightly better during the initial experiments.

### Q2: If you had to redo this week, what would you do differently?

I would validate the simulation results after each implementation step instead of waiting until the end of the week. This would make debugging easier and improve confidence in the performance evaluation. I would also allocate more time for visualization so that the comparison results could be interpreted more effectively.

## 8 MENTOR REVIEW (Mentor fills this section)

***Students submit the report by Friday 5pm. Mentor completes this section by Monday 9am before the next week begins.***

### 8A. Checklist

| Item | Status | Comment |
|---|---|---|
| Key deliverable submitted and committed to GitHub | | |
| Section 4B results table has real numbers | | |
| Section 5A has ≥ 3 candidate paper sentences | | |
| GitHub commits are clean with meaningful messages | | |
| Friday demo completed | | |
| Results are reproducible (someone else can run the code) | | |

### 8B. Technical Feedback

### 8C. Paper Progress Assessment

| Paper Section | Status After This Week | Target Week to Complete |
|---|---|---|
| Introduction | Not started / Outline / Draft / Final | Week 7 |
| Related Work | Not started / Outline / Draft / Final | Week 6 |
| Methodology | Not started / Outline / Draft / Final | Week 7 |
| Results | Not started / Data collected / Draft / Final | Week 7 |
| Conclusion | Not started / Draft / Final | Week 7 |
| References | Not started / Partial (__ refs) / Complete | Week 8 |

## 8D. Go / No-Go for Next Week

| ✅ **PROCEED** | ⚠ **FIX FIRST** | 🔴 **REPLAN** |
|---|---|---|
| All deliverables met. Continue to next week's plan. | Fix specific items before proceeding. See feedback above. | Results not credible. Stop and redesign the experiment. |

**Decision: Circle one above. If FIX FIRST — list exact items to fix before Monday:**

## 8E. Mentor Signature

| Mentor Name | Date Reviewed | Signature |
|---|---|---|
| | | |

# APPENDIX — CUMULATIVE PAPER SKELETON TRACKER

***Update this page every week. By Week 6 this should be the first draft of your paper.***

**HOW TO USE THIS TRACKER**

Each row corresponds to a paper section. Paste your best candidate sentences from Section 5A here each week.

By Week 6 you should have 3–5 sentences per row. These become the paper draft in Week 7.

Colour code: Black = confirmed content Grey = draft Red = needs rewrite

## A. Introduction

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |
| Week 7 | |
| Week 8 | |

## B. Related Work

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |

| | |
|---|---|
| Week 7 | |
| Week 8 | |

## C. Methodology

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |
| Week 7 | |
| Week 8 | |

## D. Results

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |
| Week 7 | |
| Week 8 | |

## E. Discussion / Conclusion

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |

| | |
|---|---|
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |
| Week 7 | |
| Week 8 | |

## F. References (IEEE format)

| Added in | Content / Draft Sentences |
|---|---|
| Week 1 | |
| Week 2 | |
| Week 3 | |
| Week 4 | |
| Week 5 | |
| Week 6 | |
| Week 7 | |
| Week 8 | |